

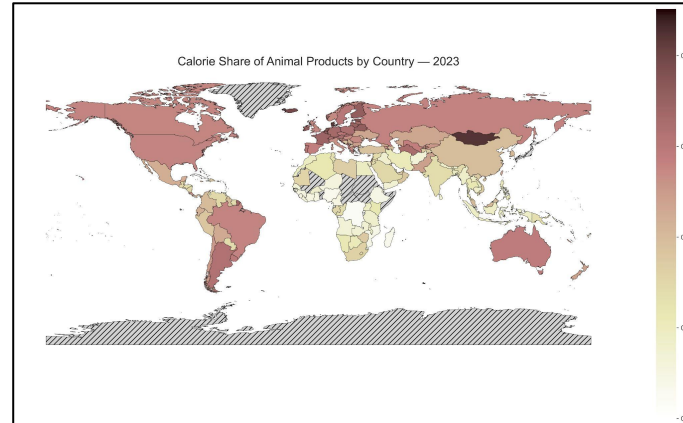
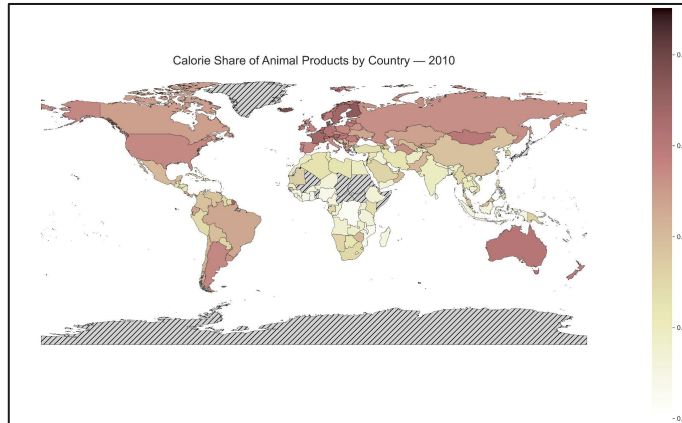
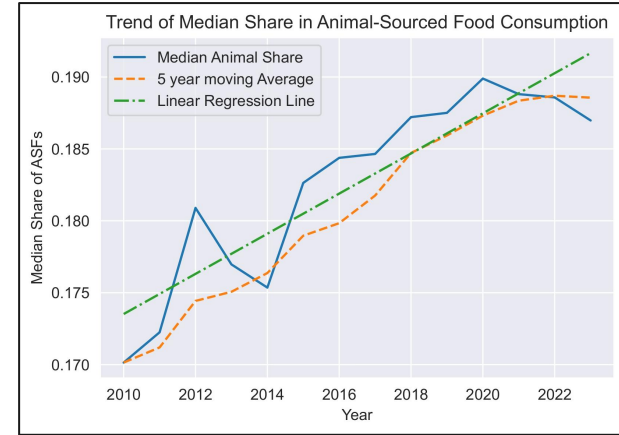


Factors contributing to Consumption of Animal-Sourced Foods

by Fabian Struensee | 114300331

Animal-Sourced Food (ASF) Share

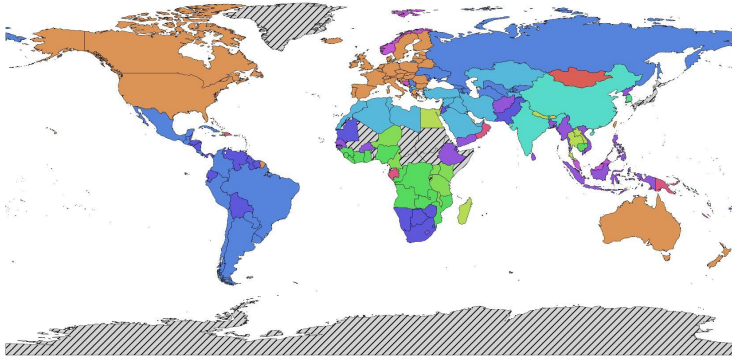
- 18.9% of kcal consumed
- 66.3% of food-related emissions (8.43 x PSF)
- Food-related emissions 25-34% of total anthropogenic GHG [1][2]
- -50% food-related emissions to maybe meet 1.5 °C goal [3]
- Other environmental and Health risks [4][5]



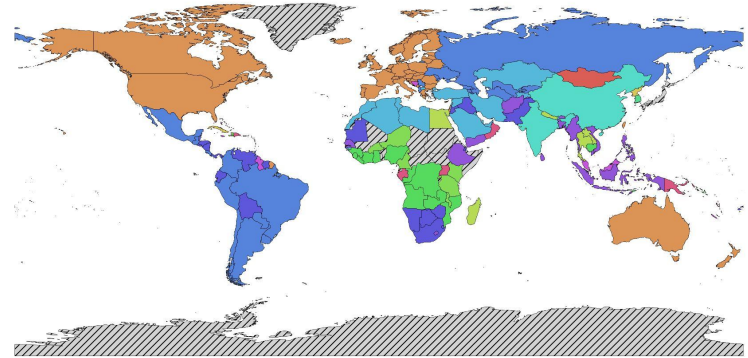
Hierarchical Clustering

- Clear regional and cultural clusters
- Barely any change over 13 years (Norway, North Korea)
- High temporal stability

Diet Clusters by Country — 2010



Diet Clusters by Country — 2023



Literature



Nutrition Transition:

- Rising incomes + urbanisation + globalisation → consumption of ASF [6][7]

Consumer Theory:

- Price + Income + Preferences → buying choice [8]

Engel's Law:

- Rising income → total spend on food rises, but share of total spending declines [9]

ASF Distribution:

- Meat consumption to income may follow a Kuznets relationship (peak at very high income) [10]

Literature Limitation

- 
- Heavy reliance on cross-sectional analyses [11]
 - Limited ability to capture persistence and long-run dynamics in diets
 - Few studies combine long time horizons with global coverage in panel frameworks [12]
 - Dietary recommendations, often based on high-income country assumptions [13]

Hypotheses



Hypothesis 1 (Income Effect):

- Higher GDP per capita is associated with a higher share of animal-sourced foods in total dietary energy consumption.

Hypothesis 2 (Urbanisation Effect):

- A higher degree of urbanisation increases ASF consumption due to shifts in lifestyles, food access, and dietary preferences.

Hypothesis 1 (Price Effect):

- Higher (relative) prices of animal-sourced foods are associated with lower consumption share.

Data

- 
- Global country-year panel dataset (2010–2023)
 - Coverage: 111 countries, 1,486 observations
 - Combines food supply, socio-economic, and price data
 - Focus on long-run structural determinants of diets

Data Sources



Dependent Variable: Share of animal-sourced foods

- Food and Agriculture Organization of the United Nations. FAOSTAT Food Balance Sheets. Accessed: 2024-12. 2024. url: <https://www.fao.org/faostat/en/#data/FBS>.

Independent Variables: Percentage urban pop, log(GDP per cap), price index

- World Bank. World Development Indicators. Accessed: 2024-12. 2024. <https://api.worldbank.org>.
- Food and Agriculture Organization of the United Nations. FAOSTAT Prices and Producer Prices. Accessed: 2024-12. 2024. <https://www.fao.org/faostat/en/#data/PP>

Control Variables: Population, aridable land per cap, trade_openess, electricity, young share

- World Bank. World Development Indicators. Accessed: 2024-12. 2024. <https://api.worldbank.org>.

Time-invariant Variables: Colonisation, religion, climate

- Centre d'Études Prospectives et d'Informations Internationales (CEPII). Colonial Relationship Database. Accessed: 2024-12. 2024. https://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=6.
- Our World in Data. Most Common Religion by Country. Accessed: 2024-12. 2024. <https://ourworldindata.org/grapher/most-common-religion>.

Methods

- 
- Fixed-effects panel regression
 - Controls for unobserved, time-invariant country characteristics
 - Time fixed effects capture global shocks
 - Standard errors clustered at country level
 - Dietary clustering used only descriptively, not in regressions

Results

Dep. Variable:		animal_share	R-squared:	0.1472	
F-statistic:		25.952	P-value	0.0000	
	Parameter	Std. Err.	P-value	Lower CI	Upper CI
Population	0.0167	0.0077	0.0317	0.0015	0.0318
arable_land_km2	0.0207	0.0114	0.0702	-0.0017	0.0431
access_electricity	-0.0059	0.0049	0.2331	-0.0156	0.0038
trade_openness	-0.0025	0.0030	0.4050	-0.0085	0.0034
log_gdp_pc	0.0502	0.0127	0.0001	0.0252	0.0751
young_share	0.0249	0.0126	0.0483	0.0002	0.0497
urban_pop_share	-0.0223	0.0173	0.1964	-0.0563	0.0116
asf_price_index	0.0010	0.0020	0.6113	-0.0028	0.0048
psf_price_index	0.0069	0.0049	0.1632	-0.0028	0.0166
const	0.1981	1.999e-17	0.0000	0.1981	0.1981

Results



Dep. Variable:	animal_share	R-squared:	0.1349
F-statistic:	30.193	P-value	0.0000

	Parameter	Std. Err.	P-value	Lower CI	Upper CI
log_gdp_pc	0.0406	0.0141	0.0040	0.0129	0.0683
log_rel_price	0.0157	0.0493	0.7509	-0.0811	0.1124
log_rel_price_x_log_gdp	-0.1496	0.4401	0.7340	-1.0130	0.7138
young_share	0.0025	0.0012	0.0401	0.0001	0.0048
urban_pop_share	-0.0011	0.0008	0.1857	-0.0028	0.0005
arable_land_km2	6.6436	4.1560	0.1102	-1.5092	14.796
log_pop	-0.0161	0.0191	0.3999	-0.0536	0.0214
const	-0.0587	0.2714	0.8287	-0.5911	0.4737

Conclusion



Hypothesis 1 (Income Effect):

- Higher GDP per capita is associated with a higher share of animal-sourced foods in total dietary energy consumption.
- Significant and supported by both models

Hypothesis 2 (Urbanisation Effect):

- A higher degree of urbanisation increases ASF consumption due to shifts in lifestyles, food access, and dietary preferences.
- Insignificant and not supported in both models

Hypothesis 1 (Price Effect):

- Higher (relative) prices of animal-sourced foods are associated with lower consumption share.
- Insignificant and not supported in both models

Sources



- [1]: J. Poore and T. Nemecek. “Reducing food’s environmental impacts through producers and consumers”. In: Science 360.6392 (2018), pp. 987–992. doi: 10.1126/science.Aaa0216
- [2]: M.Crippa et al. “Food systems are responsible for a third of global anthropogenic GHG emissions”. In: Nature Food 2.3 (2021), pp. 198–209. doi: 10.1038/s43016-021-00225-9.
- [3]: Michael A. Clark et al. “Global food system emissions could preclude achieving the 1.5° and 2°C climate change targets”. In: Science 370.6517 (2020), pp. 705–708. doi: 10.1126/science.aba7357
- [4]: WorldCancerResearchFundandAmericanInstituteForCancerResearch.Diet,Nutrition, Physical Activity and Cancer: A Global Perspective. Continuous Update Project Expert Report 2018. Third Expert Report. 2018. url: <https://dietandcancerreport.org>
- [5]: MartinC.Parlasca and Martin Qaim. “Meat Consumption and Sustainability”. In: Annual Review of Resource Economics 14.Volume 14, 2022 (2022), pp. 17–41. doi: <https://doi.org/10.1146/annurev-resource-111820-032340>
- [6]: BarryM.Popkin.“NutritionalPatternsandTransitions”.In:PopulationandDevelopment Review 19.1 (1993), pp. 138–157
- [7]: Richard York and Marcia Hill Gossard. “Cross-national meat and fish consumption: exploring the effects of modernization andecological context”. In: Ecological Economics 48.3 (2004), pp. 293–302. doi: <https://doi.org/10.1016/j.ecolecon.2003.10.009>.

Sources



[8]: Angus Deaton and John Muellbauer. Economics and Consumer Behavior. Cambridge University Press, 1980. doi: <https://doi.org/10.1017/CBO9780511805653>.

[9]: Ernst Engel. Die Productions- und Consumtionsverhältnisse des Königreichs Sachsen. Dresden: C. Heinrich, 1857.

[10]: Jennifer Rivers Cole and Suzanne McCoskey. “Does global meat consumption follow an environmental Kuznets curve?” In: Sustainability: Science, Practice and Policy 9.2 (2013), pp. 26–36. doi: 10.1080/15487733.2013.11908112.

[11]: Richard York and Marcia Hill Gossard. “Cross-national meat and fish consumption: exploring the effects of modernization and ecological context”. In: Ecological Economics 48.3 (2004), pp. 293–302. doi: <https://doi.org/10.1016/j.ecolecon.2003.10.009>

[12]: Vania Andreoli et al. “Drivers of Protein Consumption: A Cross-Country Analysis”. In: Sustainability 13.13 (2021). doi: 10.3390/su13137399.

[13]: Food and Agriculture Organization of the United Nations and World Health Organization. Sustainable Healthy Diets – Guiding Principles. Rome: FAO and WHO, 2019. url: <https://www.fao.org/3/ca6640en/ca6640en.pdf>